

A very brief introduction to Electron Swarm experiments (Enjambre de electrones)

Notas de Tim Teich

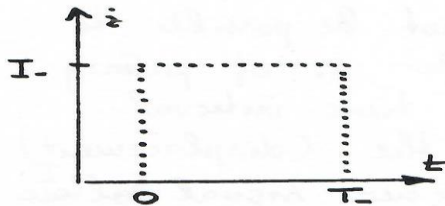
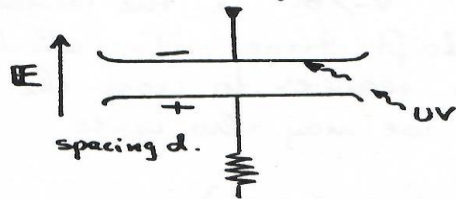
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Gas Discharges.

Introduction to collision processes with particular reference to the behaviour of electron swarms.

It is assumed that the reader is familiar with the contents of the notes "Gas Discharges: Mechanism of spark breakdown (materials science)". As in these notes, we shall confine our considerations largely to homogeneous, or initially homogeneous fields.



Let us consider a homogeneous field discharge gap. At time $t=0$ we release a large number n_0 of electrons from the surface of the negative electrode and within a very short time interval ("delta pulse of electrons"). Some or all of these electrons will then drift to the positive electrode and arrive there after a finite time interval. If we measure the current, it will look approximately as shown on the left. At time $t=T_-$ the current ceases.

We assume that the electrons undergo only elastic collisions. We can find a quantitative value of this current from energy considerations.

We assume that a constant voltage U is applied to a discharge gap of spacing d , so that the field is $E = U/d$.

The work done in moving a charge q with constant mean velocity v_- by a path length ds in the direction of field E is

$$E = \frac{U}{d}$$

v_- = mean electron velocity

$$W = F ds = nq E ds.$$

$$ds = v_- dt.$$

$$W = nq \cdot \frac{U}{d} v_- dt.$$

$$W = UI_- dt$$

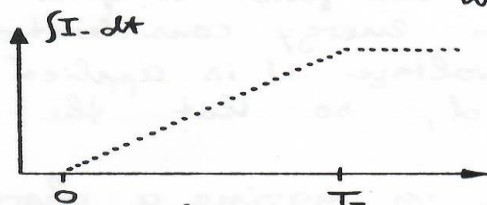
$$I_- = nq \frac{v_-}{d}.$$

$$I_-(t) = n(t)q \cdot \frac{1}{T_-}$$

$$(n(t) > 0 \text{ for } 0 \leq t \leq T_-).$$

$$I_-(t) = n(t)q \frac{1}{T_-} \quad [\text{Ec. 1}]$$

$$\{n(t) > 0 \text{ for } 0 \leq t \leq T_-\}$$



The path ds in field direction in time dt is

With $E = U/d$ we obtain

This mechanical work must equal the electrical work

and by equating we obtain

The term v_-/d is the inverse of the drift time T_- which an electron requires to cross the gap. Thus we may also write

In many cases, it may not be possible to release a sufficient number n_0 of primary electrons in a very short time interval for direct observation of the (displacement) current. To obtain a higher signal voltage for oscilloscopic recording, one integrates the current, and the observed signal will look as follows:

(input $RC \gg T_-$).

Such recordings are relatively easy to make with gases which permit a high photoelectric yield from the cathode when irradiated by an ultraviolet flash lamp.

GD 2-49)

$$v \sim \frac{E}{P}$$

(GD 2-48)

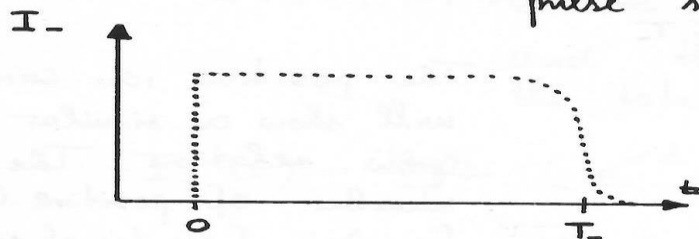
Such measurements of drift velocities (or transit times T) have been carried out by HORNBECK (and others) using the method outlined above, and by BRADBURY & NIELSE using an electric shutter method. (see GD 2-1

For many gases the investigators find that

at low and moderate values of $\frac{E}{P}$.
(See: "mobility" GD 2-48)

This indicates a charge carrier - induced dipole interaction (introd. notes p.4 ; GD 2-51/52).

In reality, the pulse shape observed is not exactly as sketched at first. If we ignore bandwidth (risetime) limitations of amplifier and oscilloscope and any deficiencies of the flash lamp, there is still a deviation from the rectangular (or ideal ramp up) pulse shape:



The "rounding off" of the top on the "tail" of the current pulse are due to (longitudinal) diffusion. The deviation from ideal pulse shape can in fact be used to determine the mean

(thermal) electron energy:

$$U \rightarrow \text{thermal velocity}; \quad \frac{1}{2} m_e U^2 = \frac{3}{2} k T_e$$

$$\nu_m \rightarrow \text{momentum transfer collision frequency}$$

$$D = \left[\frac{U^2}{3 \nu_m} \right]_{\text{AVE}}$$

(Diffusion coefficient)



$$\Gamma = - \text{grad} (D n_e)$$

while the (superimposed) diffusion particle current [Fick's law of diffusion; n_e = particle density]

So far we have assumed that the electron energies were too low to produce a detectable amount of ionization.

The number of ion pairs produced by $n_-(x)$ electrons drifting a distance dx in the direction of the field is

$$dn_+ = dn_- = n_-(x) \propto dx$$

and with $dx = v_- dt$ as before,

$$= n_-(v_- t) \propto v_- dt$$

$$\frac{dn_-}{n_-} = \propto v_- dt$$

$$\rightarrow \ln n_- \Big|_0^t = \propto v_- \int_0^t dt$$

$$\boxed{n_-(t) = n_0 e^{\propto v_- t}}$$

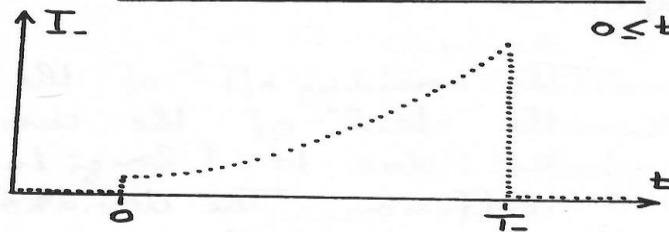
with $n_0 = n_-(0)$ (δ -pulse).

We have already calculated the (displacement) current due to $n_-(t)$ charges moving in the field — see GD-X2 & GD3-2; we have here

$$I_-(t) = n(t) q \frac{1}{T_-} \quad [\text{Ec. 1}]$$

$$I_-(t) = \frac{q \cdot n_0}{T_-} e^{\propto v_- t}$$

(time dependent electron current).



The positive ion current will show a similar growth relation. The number of positive ions formed is (see top of page)

$$dn_+ = n_-(x) \propto dx$$

and thus (see above)

$$= n_0 e^{\propto v_- t} \propto v_- dt$$

Integration gives us

$$n_+(t) = n_0 \int_0^t e^{\propto v_- t} d(\propto v_- t) = n_0 (e^{\propto v_- t} - 1)$$

If we ignore the insignificant loss of positive ions during the relatively short time interval $0 < t < T_-$ (by drifting to the cathode), we can give the positive ion current as

$$I_+(t) = \frac{n_0 q}{T_+} (e^{\alpha v_- t} - 1) \quad 0 < t < T_-$$

After $t = T_-$, no additional positive ions are formed, but an increasing number of ions will leave the discharge gap at the cathode and will therefore no longer contribute to the ion current. The maximum number of ions (and thus the maximum ion current) had been reached at $t = T_-$:

$$I_+(T_-) = \frac{n_0 q}{T_+} (e^{\alpha d} - 1) \quad \text{as } v_- T_- = d.$$

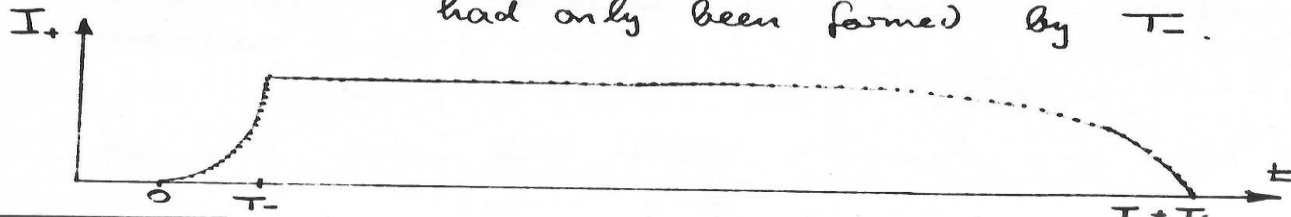
The number of ions lost from the gap up to time t is

$$\Delta n_+(t) = n_0 (e^{\alpha v_+ t} - 1)$$

and this loss has to be subtracted from the total ion current:

$$I_+(t) = \frac{n_0 q}{T_+} (e^{\alpha d} - e^{\alpha v_+ t}) \quad T_- < t < T_- + T_+$$

The time interval stated $\nearrow$ is reasonable, as the maximum number $n_0(e^{\alpha d} - 1)$ of ions had only been formed by T_- .



From the ion current pulse we can easily obtain T_+ and thus $V_+ = \frac{d}{T_+}$.

With the aid of V_+ we can then evaluate the shape of the pulse tail to yield α .

To consider steady state conditions, we substitute the δ -pulse of n_0 electrons by a steady current I_0 GD 3-3 to 4.

Classical theory of ionization: GD 3-5 to 7

α from cross sections: GD 3-8 to 10

RPD method to measure cross sections: GD 3-11 to 13.

Secondary processes and breakdown: GD 3-14 to 18/19.

Gas kinetics — summary

Energy transfer in elastic collisions

Let m_1 have kinetic energy W , collide with m_2 at rest, transferred energy ΔW .

Maximum energy transfer in central collisions:

$$\left(\frac{\Delta W}{W}\right)_{\max} = \frac{4 m_1 m_2}{(m_1 + m_2)^2}$$

$$\text{for } m_1 = m_2 : \quad = 1$$

$$m_1 \ll m_2 : \quad = 4 \frac{m_1}{m_2} \ll 1$$

$$m_1 \gg m_2 : \quad = 4 \frac{m_2}{m_1} \ll 1$$

(for derivation and discussion, see GD 2-19 to 29).

The mean value of energy transfer is lower by a factor of $\frac{1}{2}$, and the value

$$\frac{\overline{\Delta W}}{W} = \alpha = \frac{2 m_1 m_2}{(m_1 + m_2)^2}$$

is called the "accommodation coefficient".

Energy transfer in inelastic collisions

Conservation of momentum limits the amount of energy $\Delta W'$ which is available for inelastic reactions such as excitation, ionization, dissociation:

$$\left(\frac{\Delta W'}{W}\right)_{\max} = \frac{m_2}{m_1 + m_2}$$

$$\text{for } m_1 = m_2 : \quad = \frac{1}{2}$$

$$\text{for } m_1 \ll m_2 : \quad = 1$$

$$\text{for } m_1 \gg m_2 : \quad = \frac{m_2}{m_1} \ll 1$$

The mean value of energy transferred is, as with elastic collisions,

$$\overline{\Delta W'} = \frac{1}{2} \Delta W'_{\max}$$

Thus an electron can transfer its entire energy for excitation or ionization; an ion cannot transfer more than half its kinetic energy in inelastic conditions (assumed: $m_1 \approx m_2$).

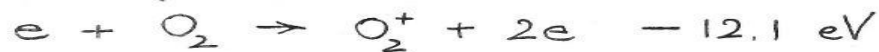
Excitation by positive ions is only of importance in very low pressure discharges where ions may well gain up to 100 eV between collisions; however, ionization by excited ions occurs over a wider range of conditions.

Ionization / dissociation.

Most ionizations are brought about by collisions with electrons; stepwise ionization may occur when charge carrier densities are high; intermediate energy levels will, in general, either be resonance levels (high excitation probability) or metastable levels (long life of excited state).

Ionization of molecules may occur in various ways (example chosen: O_2):

a) Formation of molecular ions:



b) Dissociative ionization:





c) In addition, dissociation without ionization may occur:



(the ionization potential of O is 13.6 eV, of O_2 12.1 eV).

Recombination

is the inverse to ionization / dissociation.

The simplest recombination process — recombination of one electron with one positive ion — is relatively rare; total momentum after collision must be almost equal to the momentum of the ion before collision, thus the atom can take over neither the recombination energy nor the kinetic energy of the electron, and the excess energy must be emitted as radiation. The atom may be left in an excited state, and the emission of radiation may occur stepwise, so that all series limits (of the spectra) may be observed.

Three body recombination is more likely, for instance



in this case, the remaining free electron may take over the excess energy (and thus provide conservation of momentum).

Pressure and temperature : see GD 2-1 to -4 and -12 to -14.

Velocity distribution : see GD 2-5 to -11.

Specific heat

The mean kinetic energy of particles at a temperature T is $\frac{3}{2} kT$. The particles may move in all directions in space, and the velocity components in three mutually orthogonal directions are independent of each other — one refers to these as the three degrees of freedom of translational motion. None of the degrees of freedom may be privileged, thus — on average — the energy per degree of freedom should always have the same value, and the total "inner" energy of a molecule will be

$$\overline{W_{\text{tot}}} = f \cdot \frac{1}{2} kT$$

where f the the number of degrees of freedom. The specific heat is defined by

$$C_v = \frac{d \overline{W_{\text{tot}}}}{dT},$$

and for monoatomic gases we would thus have

$$C_{v,1} = \frac{3}{2} k$$

Molecules may carry rotational energy as well; rotation may occur about three axes, and this would contribute another three degrees of freedom. However, the rotational energy with respect to the axis connecting the two atoms of a diatomic molecule is practically zero, so that we find for diatomic

molecules

$$\overline{W_{\text{rot}_2}} = kT$$

In addition to rotation, there may be oscillation, which will, with diatomic molecules, contribute a maximum value of kT ; we have $f_{\text{max}}^{\circ} = 2$ for this case as the energy will change during each oscillation cycle, from kinetic to potential energy and vice versa.

However, oscillation energy is quantized — only certain discrete values of oscillator energy are permitted. Let ν_1 be the fundamental frequency of the oscillator. Then it cannot absorb any energy while $kT \ll h\nu_1$, f° will then gradually increase with increasing temperature and will reach the maximum value $f^{\circ} = 2$ when $kT \gg h\nu_1$. The number of degrees is thus temperature dependent and not an integer, and we have

$$\overline{W_{\text{osc}}} = kT \left(\frac{h\nu_1}{kT} \right)^2 \frac{e^{-\frac{h\nu_1}{kT}}}{\left(e^{-\frac{h\nu_1}{kT}} - 1 \right)^2}$$

For the diatomic molecule we have thus

$$C_v = k \left(\frac{5}{2} \right) \xrightarrow[\text{going to}]{\text{with increasing } T} \frac{7}{2} k$$

Transport phenomena:

Diffusion: see GD 2-42 to -47.

Diffusion coefficient. $D = \frac{\sqrt{\lambda_m}}{3} = \sqrt{\frac{8k^3}{9\pi}} \frac{1}{\eta_c \sqrt{m}} \frac{T^{\frac{3}{2}}}{P}$

(because of $m^{-\frac{1}{2}}$ much greater for electrons than for ions).

Excitation / Radiation equilibrium

With thermal equilibrium, the incidence of all possible energy states is given by the BOLTZMANN distribution:

$$\frac{N_m}{N} = \frac{g_m e^{-\frac{W_m}{kT}}}{\sum_r g_r e^{-\frac{W_r}{kT}}}$$

g_m is the statistical weight of the m -th state and the nominator is the partition function, a sum over all possible excitation states of the particle. The partition function and g_m can be determined (in principle) from theory.

The expression given above can be interpreted as the excitation probability for the m -th state.

The intensity of the radiation in a spectral line which is emitted for a transition from the m -th state to the n -th state from unit volume (into 4π sr)

$m \rightarrow n$

$$S_{\nu} = N_m A_{n}^m h\nu$$

The transition probability A_{n}^m can be calculated; it is given by the inverse of the mean natural lifetime of state m against return to state n ; this lifetime is, for strong lines, of the order of 10 ns, thus we may have, typically, $A_{n}^m \approx 10^8 \text{ s}^{-1}$.

The equation given above has to be applied to all possible transitions in order to find the total intensity.

